PQ-Auth Sentinel — Research Testing Report

Dataset: unsw_nb15 | Target: attack_cat

Selected model and held-out test results

Measure	Result
Selected model	Decision Tree
Test accuracy	68.72%
Test macro-F1	0.470
Test weighted F1	0.739
Train / validation / test rows	140272 / 35069 / 82332
Input feature count	42

Algorithm comparison

Algorithm	Validation Accuracy	Validation Macro F1	Test Accuracy	Test Macro F1	Status
Logistic Regression	0.671	0.408	0.620	0.354	OK
Decision Tree	0.754	0.592	0.687	0.470	OK
Random Forest	0.749	0.565	0.676	0.472	OK
Extra Trees	0.714	0.474	0.651	0.407	OK
Soft Voting (RF + Extra Trees)	0.744	0.539	0.669	0.456	OK

Research limitation

UNSW-NB15 is a network intrusion detection benchmark. Its metrics do not directly validate ML-DSA signatures or the application's seven authentication telemetry features. Report results on each dataset separately.